UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	COMPUTER ORGANIZATION AND ARCHITECTURE
COURSE CODE	:	CSC429
EXAMINATION	:	NOVEMBER 2026
TIME	:	

MARKING SCHEME NOTES

1. This marking scheme accompanies the CSC429 mock paper (November 2026). Generated answers show working. Real past-paper questions are reproduced with their source reference; no answer is invented where the original scheme is unavailable.

2. Generated marking points state the expected answer and the allocation of marks. Real-bank questions retain their source paper and page reference.

3. Part A = 4 x 5 + 6 x 10; Part B = 1 x 20; total = 100 marks.

4. Generated slots: 2. Real-bank slots: 9. Placeholders: 0. Questions and answers are for local study use only.

5. Structure derived from 20 unique CSC429 past papers; every real question remains traceable to its source page.

FOR MARKERS' USE ONLY

This marking scheme consists of 5 printed pages

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PART A — MARKING GUIDE

QUESTION A1 (5 marks)

Topic: Parallel and multiprocessor architecture

a) List TWO (2) advantages of SMP organization.
b) Define cluster organization.
c) Give ONE (1) benefit of cluster organization.

Source: CSC429 (7).pdf, page(s) 5

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A2 (5 marks)

Topic: Instruction formats and addressing modes

Given the following Figure 2.
Instruction
Opcode
Address A
Memory
Operand
Figure 2
Label the addressing technique illustrated in Figure 2.
a)
Explain how the addressing technique in 9(a) is used.
b)
List ONE (1) advantage and ONE (1) disadvantage of the addressing technique in

Source: CSC429 (1).pdf, page(s) 5

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A3 (5 marks)

Topic: Computer generations and history

The integrated circuit is an important milestone that marks the achievement in the history of computer development.
a) In which computer generation that the integrated circuil was first introduced?
b) State the name of the inventor of the integrated circuit.
c) What are the THREE (3) main components of integrated circuits?

Source: CSC429 (20).pdf, page(s) 2

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A4 (5 marks)

Topic: Data representation and number systems

Convert -6.75 to IEEE 754 single-precision format. Give the answer in hexadecimal and show the sign, exponent and mantissa fields.

Answer: -6.7510 = 0xC0D80000 (sign=1, exponent=10000001, fraction=10110000000000000000000)

- *1 mark for the sign bit.*
- *1 mark for the biased exponent.*
- *1 mark for the 23-bit fraction.*
- *1 mark for the correct 32-bit grouping.*
- *1 mark for the final hexadecimal value.*

QUESTION A5 (10 marks)

Topic: Computer architecture and CPU organisation

Describe the specificfunction ofdifferent types of registers.

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Source: CSC429 (13).pdf, page(s) 4

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A6 (10 marks)

Topic: Input/output and interrupts

Keyboard is the most common input/output device to communicate between computer and user.

a) What is the input/output module category for keyboard?
b) State TWO (2) other devices that are in the similar input/output module category?
c) What is the basic unit of exchange for keyboard?
d) Discuss TWO (2) types of characters.

Source: CSC429 (11).pdf, page(s) 5

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A7 (10 marks)

Topic: Data representation and number systems

Answer both parts. Show your complete workflow.

a) Convert -11.375 to IEEE 754 single precision. Give the answer in hexadecimal.

Answer: -11.37510 = 0xC1360000 (sign=1, exponent=10000010, fraction=01101100000000000000000)

- *1 mark for sign, exponent and mantissa.*
- *1 mark for the biased exponent.*
- *1 mark for the 23-bit fraction.*
- *1 mark for the 32-bit binary grouping.*
- *1 mark for the hexadecimal result.*

b) Convert the IEEE 754 single-precision value 0x40500000 to decimal.

Answer: 0x40500000 = 3.2510

- *1 mark for the sign bit.*
- *1 mark for the exponent field.*
- *1 mark for the fraction field.*
- *1 mark for applying the bias and significand.*
- *1 mark for the final decimal value.*

QUESTION A8 (10 marks)

Topic: Input/output and interrupts

a) Define interrupt.
b) Describe TWO (2) different types of interrupt.
c) Discuss ONE (1) advantage and ONE (1) disadvantage of interrupt

Source: CSC429 (3).pdf, page(s) 4

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A9 (10 marks)

Topic: Computer architecture and CPU organisation

Describe the function of each register in the computer system.

Source: CSC429 (14).pdf, page(s) 3

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

QUESTION A10 (10 marks)

Topic: Memory hierarchy and cache

Question 7 is based on the following scenarios.
While cleaning his house, Ahmad found an old cassette tape of a popular 80's band. He played the tape and listened to several songs according to the listed tracks.

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Hafiz was tasked to research on the history of Malay culture by his professor. He went

to the National Archive to search for information and manage to access one the resources which was stored in a floppy disk.

Sara is planning to buy a new computer. She aims to get one with good processing speed. While researching for computer specifications, she found out that larger cache space has a direct effect towards processing performance.

Hana loves playing role-playing games (RPG) and is planning to buy a newly released game that requires a high-speed RAM. But, she is devastated when she found out that her game console does not support the minimum RAM requirement of the game.

a)
Based on the above scenarios, identify the access method for each storage devices/memory mentioned in (i), (ii), (iii) and (iv).

b)
Describe SRAM and DRAM.

c) Which level of cache is the fastest in a three-level cache architecture? Which type of RAM is used to implement the cache?

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Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

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PART B — MARKING GUIDE

QUESTION B1 (20 marks)

Topic: Assembly-language programming

Question 1 is based on Figure 2 and Table 1.

y = A -
2C
* D
B + A2

Figure 2

Write an appropriate program using correct assembly language notations to compute the equation shown in Figure 2 using instructions for one-address, two-address, and three-address. The available instructions are provided in Table 1. The register labeled A, B, C, and D are fixed, while registers x, y, and z can be used temporarily. The final answer must be stored in register y.

Table 1 : Machine instruction

Three-address	One-address	Two-address
LOADp		
MOV (p,q)		
ADD (p.q.r)		
STORp		
ADD (p,q)		
SUB (p,q,r)		
ADDp		
SUB (p,q)		
MUL (p.q.r)		
SUBp		
MUL (p,q)		
DIV(p,q,r)		
MULp		
DIV(p,q)		
DIVp		

Source: CSC429 (18).pdf, page(s) 6

Answer: The supplied source pack does not include a marking scheme for this past-paper question. Use the source paper and its mark allocation; no answer has been invented.

END OF MARKING SCHEME

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